

CreepPFN: A Material-Conditioned Prior-Data Fitted Network for Concrete Creep Forecasting

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Abstract

Forecasting concrete creep from short, irregular measurement histories requires a model that accounts for material variation and limited experimental data. This study introduces CreepPFN, a material-conditioned prior-data fitted network for predicting later compliance increments. A generator calibrated exclusively on training sources links material descriptors to distributions of curve magnitude and shape. Synthetic tasks pretrain an attention network, which is subsequently fine-tuned on measured training curves. Three trained members combine material properties and available readings through day 10 to forecast at requested future times without specimen-specific retraining. Five literature-source-disjoint folds evaluated 610 curves from 65 sources in the Northwestern University (NU) database. The revised ensemble achieved 17.40% source-macro normalized mean absolute error (NMAE), with a 95% source-bootstrap interval of 14.96–20.14%, compared with 26.51% for a Kelvin prefix baseline. Nominal 90% marginal predictive intervals achieved 94.51% source-macro coverage. A bounded validation search retained four encoder layers, four attention heads and width 256. Architecture ablations indicated contributions from the combined attention paths and the time gap between the last observation and each forecast. Revised retraining did not improve the historical ensemble’s 16.80% NMAE. Frozen-model evaluations yielded mean fold curve-macro NMAE of 10.39% on 41 King Mongkut’s University of Technology Thonburi (KMUTT) laboratory curves and 20.71% on 25 other-literature curves. These external results depend on assumed loading stress and an unverified modulus-age proxy. Previously examined evaluation cohorts were reused, so the revised findings remain exploratory. The contribution is the integration of material-linked synthetic task generation with early-measurement-conditioned creep forecasting. Independent-source validation and verified experimental metadata are required to establish reliability for practical application.

Keywords: Concrete creep; prior-data fitted network; synthetic training; material properties; attention; uncertainty.

1 Introduction

Concrete continues to deform under sustained load. This time-dependent response, called creep, contributes to the long-term deformation of concrete structures. Its development depends on material composition, loading age and environmental conditions (Wan-Wendner et al., 2015). Reliable forecasting must therefore account for differences between concretes and between testing conditions. Early measurements provide additional information about the response of an individual specimen. Combining those measurements with material properties offers a practical basis for predicting later behaviour before a complete creep test is available.

The main learning challenge is limited experimental diversity rather than merely a small number of recorded values. Repeated measurements describe the evolution of existing specimens; they do not introduce new materials or testing conditions. Experimental records also differ in duration, sampling frequency and available descriptors. The Northwestern University (NU) concrete creep database documents this heterogeneous evidence base and the importance of consistent recording (Hubler et al., 2015). A model fitted directly to pooled readings can therefore learn densely represented experiments more readily than sparsely represented material conditions. Forecasting a specimen from a new source requires more than fitting the curves already available.

This task also requires information beyond material properties alone. Specimens with similar recorded strength or stiffness may follow different measured histories. Early readings help distinguish their responses, but provide limited evidence about later deformation. The methodological need is to learn plausible continuations from this short history despite uneven training coverage. The evaluation must then distinguish

this forecasting capability from access to related specimens or later measurements. These requirements motivate a material-informed training distribution together with an explicit separation of experimental sources.

Existing research provides several foundations for this task. Constitutive models such as Model B4 relate creep and shrinkage to composition and testing conditions (Wan-Wendner et al., 2015). Updating methods incorporate early measurements to revise predictions and their uncertainty (Sousa et al., 2018). Data-driven approaches learn relationships directly from experimental records. Tree ensembles have been used for concrete compliance prediction (Liang et al., 2022). Recurrent models with attention combine historical measurements with material and environmental inputs (Fan et al., 2024). Creepformer uses a convolutional network and a transformer to forecast from early creep observations (Li et al., 2025). The Triple Attention model incorporates density, strength and elastic modulus in time-dependent prediction (Dokdua et al., 2025). Probabilistic creep models have also combined constitutive information with neural networks and Gaussian-process uncertainty estimates (Tao et al., 2026).

These studies establish that early measurements, material descriptors and uncertainty estimates are all relevant to creep forecasting. They also show that attention-based prediction is already part of this field. Their methods use different response definitions, input variables and evaluation protocols. Published accuracy values therefore cannot establish which approach performs best on a common early-to-later forecasting task. In particular, an evaluation should distinguish learning within familiar experimental sources from forecasting curves reported by withheld sources. Otherwise, shared testing conditions can make transfer to a new source difficult to assess.

Prior-data fitted networks (PFNs) offer a way to learn prediction rules across generated tasks before applying them to a new dataset (Müller et al., 2022). TabPFN learns from synthetic tabular datasets and uses labelled examples as context for new predictions (Hollmann et al., 2025). TabICL extends synthetic pretraining and in-context learning to larger tabular classification datasets (Qu et al., 2025). Their relevance is the separation of learning across tasks from prediction within a new task. This suggests a way to expose a creep predictor to more varied forecasting situations than the measured curves alone provide. TabPFN-TS extends tabular PFN prediction to time series (Hoo et al., 2026), while ApolloPFN uses temporal task generation and external covariates (Potapczynski et al., 2026). These general methods motivate the learning strategy, but do not specify a material-conditioned distribution of concrete creep curves.

The central difficulty is that many recorded time points do not provide equally many independent examples of concrete behaviour. The present NU evaluation contains 5,646 later readings, but only 610 eligible curves from 65 literature sources. Source sizes range from one to 58 curves, and 24 sources have no recorded modulus (Section 4). Direct training must learn material relationships, temporal evolution and responses to irregular sampling from this limited and uneven collection. Without appropriate weighting, pooled observations can give disproportionate influence to densely sampled curves or large experimental programmes. Missingness patterns may also identify sources rather than transferable material behaviour. A larger network alone does not supply the missing experimental diversity.

Evaluation can conceal this difficulty if the data hierarchy is ignored. A random split of readings can place observations from the same specimen in both training and testing. Even a curve-level split can retain closely related experiments from the same publication on both sides. Such overlap does not establish transfer to an unseen source. Using later responses to construct input features or calibrate a synthetic generator would additionally leak information unavailable at the forecast cutoff. Direct training is not inherently leaky, and synthetic training does not automatically prevent leakage. Both require source separation and training-only preprocessing. Moreover, a new source may occupy a different material or measurement distribution from the training sources, so within-source accuracy alone cannot establish the intended forecasting capability.

The unresolved need is therefore a learning framework that represents plausible variation beyond the observed training curves while retaining their material relationships and the new specimen’s measured history. Generic tabular pretraining demonstrates the potential of learning across synthetic tasks (Hollmann et al., 2025, Qu et al., 2025); applying this principle to creep requires a defensible distribution of curve magnitudes, shapes and observation schedules. CreepPFN investigates a material-conditioned task generator calibrated only on training sources, coupled to prediction from irregular early readings. The intended benefit is broader exposure to plausible forecasting tasks under limited experimental diversity. Synthetic tasks do not create new independent evidence or guarantee transfer outside the calibrated distribution. A matched real-only training control remains necessary to isolate the benefit of synthetic pretraining.

CreepPFN addresses this gap through three connected components. First, a generator links density, com-

pressive strength, recorded modulus and first-reading time to fitted curve magnitude and shape. Sampling from that generator creates varied practice tasks with irregular observation times. Second, an attention predictor combines material properties with early readings to forecast directly at requested future times. Third, measured training curves adapt the pretrained predictor, and three independently trained members provide the final forecast and predictive interval. The proposed novelty lies in this concrete-specific formulation and its evaluation. It does not rest on a claim that attention, early-measurement updating or temporal PFNs are individually new.

The objective is to assess whether this formulation can forecast additional recorded compliance from measurements through day 10. The experiments measure prediction error and interval coverage across NU literature-source groups. A bounded hyperparameter search examines model settings. Retrained component removals assess the contribution of attention, material conditioning, query information and predictive scale under those settings. Evaluations with fixed model weights use laboratory curves from King Mongkut’s University of Technology Thonburi (KMUTT), Thailand, and a separately labelled literature set. These evaluations examine performance beyond the NU cohort. Their interpretation remains conditional on response-conversion and provenance assumptions.

Data leakage occurs when information reserved for evaluation enters model development. To limit this risk, each NU fold separates training, validation and test literature-source groups. The synthetic generator, feature scaling and missing-value imputation are fitted only on training sources. Validation sources select hyperparameters and checkpoints. External curves do not enter these fitting or selection procedures. At prediction time, a test specimen contributes only its material inputs, early readings and requested query times; later response values are reserved for scoring. However, the revised architecture assessment follows inspection of earlier results on the same test partitions. This reuse creates a risk of indirect selection bias even without test-label access during fitting. The revised results are therefore exploratory rather than independent confirmation.

Section 2 provides the detailed literature comparison. Section 3 introduces the overall concept and model architecture through an explanatory figure. Section 4 presents data sources and graphical characteristics. Section 5 defines the training and evaluation protocol. Section 6 reports prediction results and ablation studies. Section 7 explains the application workflow and external examples. Section 8 interprets the results, limitations and priorities for further research. The final section presents the conclusions. Appendices retain the mathematical specifications and detailed supporting tables.

2 Related work

2.1 Concrete creep models and updating

Constitutive models provide physically motivated long-term predictions. Model B4 relates creep and shrinkage to concrete composition, strength and testing conditions. Its calibration uses laboratory and bridge data and addresses uncertainty and Bayesian updating (Wan-Wendner et al., 2015). Bayesian updating revises an earlier predictive distribution using new measurements. Its use for creep forecasts from early specimen measurements has also been studied directly (Sousa et al., 2018). These models remain important engineering references. The available core descriptors in this study do not contain the composition and environment fields needed for a matched B4 implementation. Accordingly, the present task concerns forecasting from a limited descriptor set and an early measured prefix; it is not a test of superiority over B4 or the first use of early measurements for probabilistic creep prediction.

Data-driven concrete models have addressed several related targets. Liang et al. fitted tree ensembles to Northwestern University compliance observations with mixture, loading and environmental inputs (Liang et al., 2022). Fan et al. combined gated recurrent units and self-attention for creep and shrinkage forecasting using material, environmental and historical inputs (Fan et al., 2024). Creepformer uses early creep observations with a CNN–transformer and a data-completion procedure for variable sampling intervals (Li et al., 2025). An earlier Triple Attention model by the same authors uses density, strength and elastic modulus in an autoregressive transformer for time-dependent creep prediction (Dokdua et al., 2025). Tao et al. combined a B4-informed neural network with Gaussian-process uncertainty estimates for creep prediction (Tao et al., 2026). These studies establish early-reading forecasts, material-aware attention and probabilistic creep prediction as prior work. Their reported scores do not estimate accuracy for the present day-10 increment target and literature-source-disjoint test protocol.

2.2 Prior-data fitted forecasting

Prior-data fitted networks (PFNs) learn from tasks sampled from a specified prior (Müller et al., 2022). TabPFN demonstrates this strategy for small tabular datasets (Hollmann et al., 2025). TabICL uses synthetic pretraining with column-then-row representations and a subsequent in-context transformer for tabular classification (Qu et al., 2025). The cited TabICL study addresses classification and scalability, rather than creep regression. Both studies motivate learning across tasks; their results do not establish that a generic synthetic prior covers concrete response regimes. TabPFN-TS converts temporal observations into tabular regression features for forecasting (Hoo et al., 2026). ApolloPFN already uses synthetic temporal priors and external covariates (Potapczynski et al., 2026). The remaining domain requirement is to link measured material descriptors to plausible curve variation and preserve source separation when calibrating that prior.

Table 1: Comparison with the closest method families. The cited studies use different response targets, feature sets, datasets and metrics. Their published scores are therefore not direct benchmarks for the day-10 source-disjoint task. The final row reports this study’s protocol.

Study	Predictor information and target	Learning strategy	Evaluation and relation to CreepPFN
Model B4 (Wan-Wendner et al., 2015)	Composition, strength and conditions; long-term creep and shrinkage.	Constitutive form calibrated with laboratory and bridge records.	Physically motivated reference; required condition fields are absent from this cohort.
Sousa et al. (Sousa et al., 2018)	In-situ measurements; future creep deformation.	Bayesian updating of prior model distributions.	Monitored-structure setting; early-measurement uncertainty updating predates CreepPFN.
Liang et al. (Liang et al., 2022)	NU properties, loading and time; compliance observations.	Random forest and gradient-boosted tree ensembles.	Train/test compliance regression; different input and forecast task.
Fan et al. (Fan et al., 2024)	Material, environment and history; creep and shrinkage.	Gated recurrent units with self-attention; recursive multi-step forecasts.	NU and field tests; different target, horizon and protocol.
Creepformer (Li et al., 2025)	Early creep history without material properties; later creep.	CNN–transformer with random-forest data completion.	NU and Yihui evaluations; early-reading forecast precedes this study.
Triple Attention (Dokdua et al., 2025)	Density, strength, modulus and standardized daily history; creep.	Autoregressive transformer with temporal, feature and batch attention.	Earlier work by these authors; material-aware creep attention is established.
Tao et al. (Tao et al., 2026)	Creep records and B4 model information; creep predictions.	B4-informed neural network with Gaussian-process uncertainty.	NU and laboratory evaluations; probabilistic creep prediction is established.
TabPFN-TS (Hoo et al., 2026)	Temporal features and covariates; general time-series forecasts.	Pretrained tabular PFN for time-series regression.	General forecasting benchmarks; no fold-calibrated concrete prior.
ApolloPFN (Potapczynski et al., 2026)	Time history and external covariates; general forecasts.	Time-aware PFN with a synthetic temporal prior.	Exogenous forecasting benchmarks; covariate-aware temporal PFNs precede this study.
CreepPFN (this study)	Day-10 irregular prefix, density, strength and recorded modulus; later compliance increments.	Fold-calibrated material-to-curve prior, synthetic-task pretraining and real-source adaptation.	Five source-disjoint NU folds; 610 eligible curves and source-macro NMAE. Separate frozen-model external test.

2.3 Research gap and contribution

The methodological gap arises from the mismatch between the forecasting task and the available evidence. The predictor must extrapolate from a short, irregular history, whereas the training evidence consists of a limited number of heterogeneous and unequally represented experimental sources. Direct fitting can exploit these records, but additional readings from familiar curves do not resolve limited coverage of new

materials and source conditions. Existing creep predictors establish the value of history and material inputs. TabPFN and TabICL establish synthetic task pretraining as a viable learning strategy in tabular settings (Hollmann et al., 2025, Qu et al., 2025). Neither result alone specifies how to construct material-dependent creep tasks without incorporating withheld experiments.

CreepPFN addresses this gap by estimating a distribution of curve parameters conditional on recorded training-source properties. Sampling generates varied irregular history-to-future tasks, and real training curves subsequently adapt the predictor. Source-disjoint evaluation assesses the resulting forecasts beyond the sources used to fit the generator and network. This design separates the proposal to broaden training-task variation from the requirement to protect evaluation data. It does not establish that synthetic pretraining is superior to all forms of direct real-data training or eliminates distribution shift.

The novelty claim concerns the integrated method and evaluation design. It does not imply that each component is new. Direct comparisons with Creepformer, Triple Attention and pretrained temporal PFNs under the same protocol remain absent. The literature table describes methodological differences rather than measured superiority.

3 Model architecture and overall concept

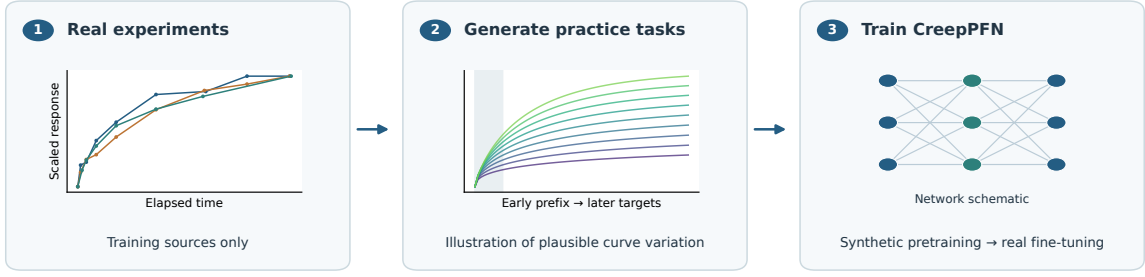
3.1 From practice curves to a specimen forecast

Figure 1 summarizes CreepPFN in two stages. During development, a generator creates plausible creep curves from relationships observed in real training experiments. The network repeatedly sees the beginning of a generated curve and learns to predict its later readings. Real training curves then adjust the learned predictor. Learning from generated tasks is called pretraining. Adjustment using real training curves is called fine-tuning. During application, the trained model receives the early measurements and properties of a new specimen.

In this paper, a *prior* means the rule used to generate plausible curves before the new specimen is observed. The rule permits variation in response magnitude, curve shape and measurement noise. A *prior-data fitted network* learns from tasks sampled using that rule. Understanding the workflow does not require Bayesian calculations. The generator is an empirical approximation, and the predicted distributions are not claimed to be an exact Bayesian posterior.

A TRAINING

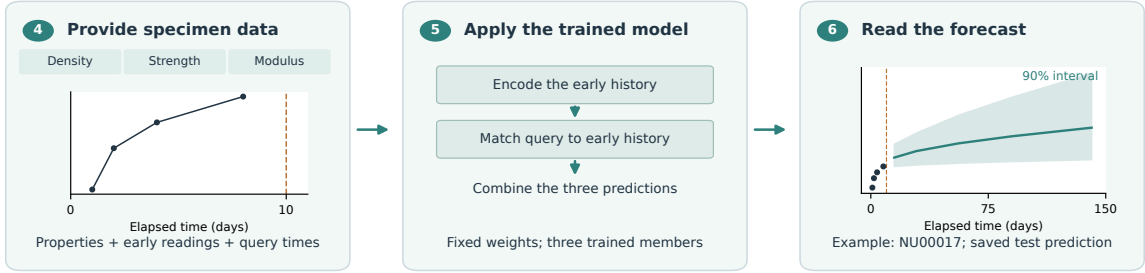
Learn from material-linked practice curves



Prior = a rule for generating plausible curves from training-source material relationships.

B APPLICATION

Use early readings to forecast a specimen



Evaluation: later measured responses are held back and compared with the completed forecast.

Figure 1: Illustrated CreepPFN workflow linked to Sections 3.3–3.5. Stages 1–2 show how real training-source records calibrate the material-conditioned generator described in Section 3.3. Stage 3 shows synthetic pretraining followed by real-data fine-tuning. Stages 4–5 summarize the specimen inputs and attention-based processing described in Section 3.4. Stage 6 shows the saved NU00017 forecast discussed in Section 3.5. The measured training curves are scaled by their individual maximum absolute increments for display. Practice curves and the network symbol are schematic illustrations. The green line and shading in Stage 6 are the saved point forecast and central 90% marginal interval; the dashed line marks day 10.

3.2 Inputs and prediction target

The inputs are density ρ , compressive strength f_c , recorded 28-day elastic modulus E_{28} , early measurement times and early responses. Requested future times are also supplied. An explicit flag identifies missing modulus. The first-reading elapsed day, t_a , is included because some experiments begin recording after loading. This quantity differs from concrete age at loading.

Compliance describes strain per unit stress. The model predicts its additional recorded amount after the first reading:

$$y_i(t) = J_i^{\text{obs}}(t) - J_i^{\text{obs}}(t_{a,i}). \quad (1)$$

Here J_i^{obs} is recorded compliance, measured in $\mu\epsilon/\text{MPa}$. The target is therefore the increase relative to the first measurement. It does not require an elastic correction inferred from the complete future curve.

The early readings are called the *context* or *prefix*. The *early history* comprises their measurement times and compliance increments. The evaluation uses all available readings through day 10 as context. The *anchor* is the first-reading time t_a ; the compliance measured at that time defines the zero reference in Eq. (1). It is neither the day-10 cutoff nor the concrete age at loading.

For illustration, consider a specimen measured on elapsed days 1, 3 and 7 before the cutoff. Its early history contains the three time-increment pairs, and its anchor is $t_a = 1$ day. Each increment is calculated by subtracting the day-1 compliance, so the first increment is zero. A reading exactly on day 10 is not required. A *query* is a future time at which a forecast is requested. The model predicts directly at each query. It does not need a daily grid or recursive predictions between observations.

3.3 Linking synthetic curves to material properties

Stages 1 and 2 in Figure 1 show how measured experiments become synthetic practice tasks. Stage 1 contains real curves and their material records from the training sources. Stage 2 shows the range of plausible curve shapes produced by the material-conditioned generator. These illustrated curves explain the role of the generator; they are not additional experiments or the exact curves used to report model accuracy.

The generator first fits two simple curve families to complete real training curves. Both describe increasing responses that gradually flatten. One family uses a Weibull form. The other combines responses with several retardation times and a logarithmic term. Their mathematical definitions appear in Appendix A.

The fitted curves have parameters controlling magnitude and shape. Regression links these parameters to density, strength, modulus and the first-reading time. Residual variation between sources and between curves allows specimens with similar recorded properties to behave differently. A generated task combines a training-source property record, an irregular measurement schedule and sampled curve parameters. Added reading noise represents part of the discrepancy between measured and fitted responses.

This mechanism links synthetic data to measured material relationships. It does not assign one fixed creep curve to each material. Instead, the material properties shift the distribution from which magnitude and shape are sampled. The relationships may also reflect unrecorded conditions such as humidity and mixture composition. Generated tasks increase the variety of training examples; they do not increase the number of independent experimental sources. Stage 3 in Figure 1 then uses these generated tasks for pretraining before real-curve fine-tuning.

3.4 How the network uses early measurements

Stages 3–5 in Figure 1 connect training to application. Stage 3 represents model training. Stage 4 shows the information supplied for a new specimen: density, strength, modulus, early readings and requested future times. Stage 5 summarizes the two main calculations. The network first encodes the measured history and then matches each future query to that history.

The network converts each early time–response pair into a numerical representation called a token. A separate small network encodes the material descriptors. This property representation is added to the measurement tokens. The context encoder then processes the tokens through four attention layers.

Attention assigns learned weights to relationships between measurements. In the context encoder, this lets a reading be interpreted together with the surrounding measured history. For each future query, a second attention operation compares that query with the encoded history. The query includes the time since the last early reading, which describes the extrapolation gap. These learned weights are computational operations rather than causal explanations of material behaviour.

The selected full architecture uses width 256, four encoder layers and four attention heads. Width is the size of the internal numerical representation. Four heads allow four learned comparisons within each attention operation. The feed-forward layers have width 768. Each member contains 3,099,906 trainable parameters. The detailed input transformations and forward equations are given in Appendix B.

3.5 Reading the forecast and uncertainty range

Stage 6 in Figure 1 shows how to read the output. Black points are the early measurements supplied to the model. The dashed line marks day 10, when the observation window ends. The green line is the point forecast at requested later times. The shaded region is the central 90% marginal predictive interval. The panel uses the saved test prediction for NU00017 rather than a schematic result.

Each member predicts a central response and a distribution of plausible responses at each query time. The final point forecast averages the response medians from three independently trained members. The shaded range contains the middle 90% of their combined predictive distribution. Combining members in this way is called an ensemble.

A wider range indicates greater predicted uncertainty at that time. A 90% interval is not a guarantee that the whole future curve lies inside the shading. Its empirical coverage must be checked against later measurements. The output does not enforce monotonicity, nonnegative increments or a joint distribution over the complete trajectory. Appendix C gives the training objective and interval calculation.

4 Data sources and characteristics

4.1 Available experiments and their roles

The main cohort is a curated extract of the Northwestern University (NU) concrete creep database (Hubler et al., 2015). It contains 617 curves from 65 literature-source groups and 9,790 retained readings through day 160. NU supplies the training, validation and test partitions. A literature source is the grouping unit. Different publications do not necessarily represent independent physical laboratories.

The external cohorts contain 41 curves from the Concrete and Materials Testing Laboratory at KMUTT, Thailand, and 25 curves labelled as other literature. The KMUTT records are labelled as ASTM C512 compression-creep tests (ASTM International, 2015). The source inventory lists experiments by Rossi et al., Choi et al., Hong et al. and Zhao et al. (Rossi et al., 2013, Choi et al., 2025, Hong et al., 2024, Zhao et al., 2016). However, individual articles cannot be mapped to the 25 literature rows. Those rows are therefore treated as one unresolved literature-labelled set.

Table 2: Origins in the local 683-curve metadata snapshot. Only NU readings contribute to prior calibration, fine-tuning and model selection. The 41 KMUTT and 25 other-literature curves supply a separate external evaluation with frozen NU-trained models. The 25 literature curves have no individual publication identifiers in the metadata.

Origin	Curves	Provenance	CreepPFN use
NU database	617	Published creep experiments in 65 literature-source groups (Hubler et al., 2015)	Prior fitting, adaptation, validation and five-fold testing
KMUTT concrete laboratory	41	Rows labelled as ASTM C512 tests (ASTM International, 2015)	External day-10 test only
Other literature set	25	Generic literature label; article mapping unavailable	External day-10 test only

4.2 Material variation and measurement schedules

Figure 2 summarizes the available curves. NU is much larger than either external cohort. The density and strength panels express each bin as a percentage of its own cohort. This allows distributions to be compared without making the smaller cohorts disappear behind the NU counts. These descriptive graphs show recorded variation; they do not establish independent sampling across material classes.

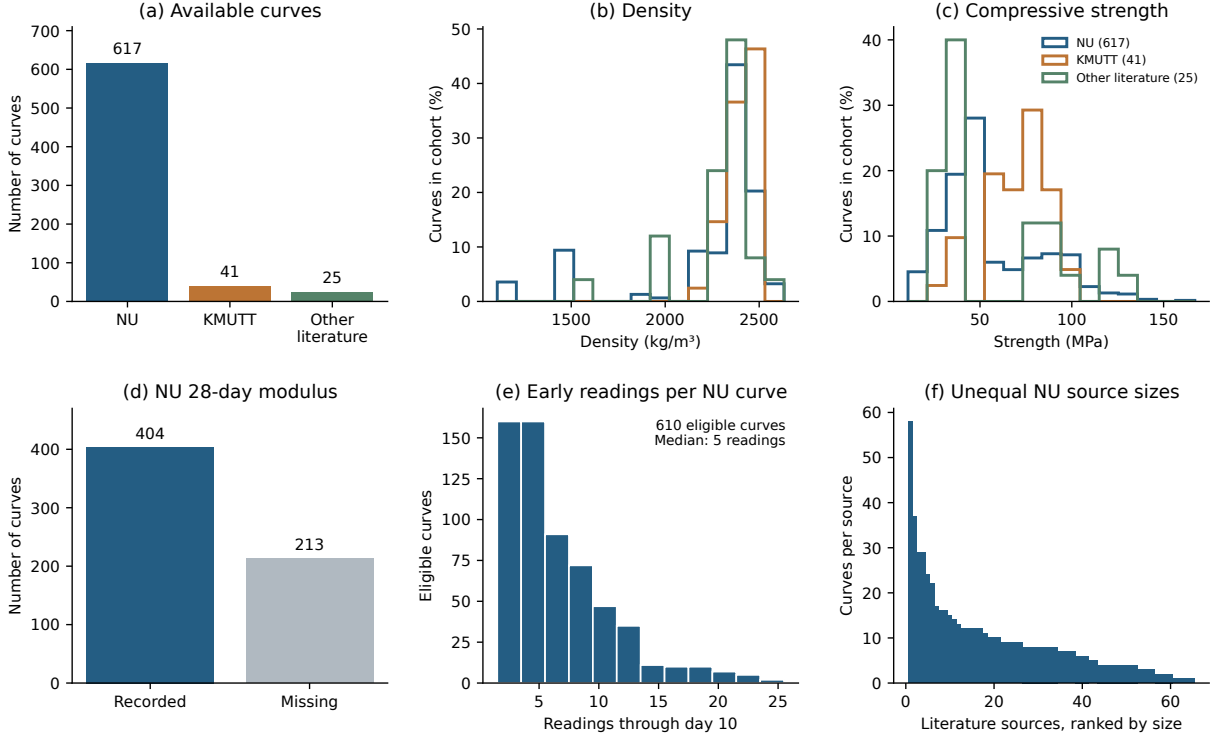


Figure 2: Characteristics calculated from actual curve records. Panels (a)–(c) include NU, KMUTT and the other-literature set. Density and strength histograms use common bins and normalize within each cohort. Panels (d)–(f) describe NU modulus availability, early-reading counts for the 610 eligible curves, and the 65 source sizes. Curves are counted once, without duplicating them across folds.

NU density ranges from 1106.4 to 2632.35 kg/m³. Compressive strength ranges from 10.8 to 167.0 MPa. Recorded E_{28} is available for 404 curves and missing for 213 curves (34.5%). Twenty-four sources have no recorded modulus for any curve. Modulus availability is therefore associated with source membership. Its missingness cannot be assumed random.

The typical eligible curve has five early readings and seven later readings. Sampling is irregular. The median gap between adjacent NU readings is four days, while the largest gap is 113 days. Source sizes range from one to 58 curves. This imbalance motivates equal weighting of sources in evaluation.

The exported descriptors do not include humidity, mixture proportions, loading stress or specimen geometry. Their effects cannot be separated from recorded material associations. Measured curves also contain local decreases: 187 curves have at least one. The synthetic families describe smooth increasing latent responses, with reading noise added around them.

4.3 Data preparation and external response conversion

For NU, flagged observations are removed and duplicate times are averaged. Readings outside days 0–160 are omitted. Import requires at least four retained times, a first reading by day 10, a final reading at or after day 28, a source identifier and positive density and strength. The inherited extract also used descriptor, duration and full-curve fit-quality filters. This curation can favour smoother or better-recorded experiments.

At evaluation, each curve needs at least two readings through day 10 and at least one later reading. This leaves 610 NU curves from all 65 sources, with 5,646 later targets. The other seven curves remain in the import inventory but cannot support the specified forecast task. Original later readings supply the evaluation targets. Daily interpolation and full-curve fits do not create test labels.

The supplementary workbook contains all 66 external curves and 1,195 readings. Density, converted strength and converted modulus are checked against metadata to align the rows. The curves provide 350 early readings and 845 later readings through day 156. All meet the day-10 evaluation rule.

External strain is reported in microstrain. Strength and modulus are converted from kgf/cm² to MPa using a factor of 0.0980665. A nominal stress of $0.4f_c$ converts strain to compliance before first-reading

subtraction. This follows the local data convention. Curve-specific loading stress is unresolved for the literature set, so equivalent response scaling is not verified.

The external recorded modulus, E_{rec} , is supplied as a proxy for the model’s E_{28} input. Its measurement age is unknown. A sensitivity evaluation marks that input missing and applies the NU training-fold imputation instead. The first external reading is at elapsed day zero; this does not specify concrete age at loading. External results remain conditional on these data assumptions.

5 Experimental design

5.1 Separation of experimental sources

Curve IDs are sorted before five-fold `GroupKFold` with literature source as the group. Each of the 617 imported curves enters exactly one outer test fold. Within each outer development partition, a seeded permutation of its source groups assigns approximately 20% of groups to validation. Remaining groups calibrate a new prior, feature scaler and imputation statistics for that fold and supply measured futures for fine-tuning. Training, validation and test groups are disjoint within a fold. Seven assigned curves do not meet the day-10 context-and-future rule, leaving a common evaluable set of 610 curves, all 65 sources and 5,646 future readings. Every method uses this same set. Validation selects each fitted checkpoint without gradients or labels from that fold’s test partition. The revised architecture choice uses saved validation scores, but it follows earlier inspection of test results on the same partitions. Its reassessment therefore cannot be interpreted as a new independent outer test.

Table 3: Fold composition from the real imported cohort. Each train, validation and test cell is curves/sources. Eligible test curves have at least two readings through day 10 and one later reading. Later readings are original recorded test targets through day 160. The fold copies do not add independent curves.

Outer fold	Train	Validation	Test	Eligible test curves	Later readings
1	402/42	91/11	124/12	124	918
2	365/41	128/10	124/14	124	1,036
3	395/42	99/10	123/13	122	1,019
4	382/42	112/10	123/13	120	1,648
5	382/42	112/10	123/13	120	1,025
Outer-test union	—	—	617/65	610	5,646

The test partitions contain 12–14 literature sources each. Their eligible curves are similar in number, but later-reading counts range from 918 to 1,648. Consequently, pooling all readings without source weighting would emphasize sources with dense future schedules. The reported primary metric instead averages curve errors within a source and then gives each source equal weight. Model training and checkpoint selection use only the corresponding development sources. Test targets remain at their recorded irregular times; no daily interpolation or full-curve analytic fit supplies evaluation labels.

5.2 A common day-10 forecast task

Day 10 defines an early testing scenario. It provides at least two context readings for 610 imported NU curves and gives every method the same observation window. It is not a fitted physical parameter. No alternative forecast origin was evaluated, so the experiments do not establish that day 10 is optimal. Synthetic pretraining uses varied prefix lengths; real testing fixes this common day-10 task.

5.3 Training, validation selection and ensemble

Four historical pretraining configurations share the same fold prior and 5,000 generated tasks per epoch. The full width-256, four-layer configuration generates a new set of synthetic examples each epoch; pre-training uses AdamW (Loshchilov and Hutter, 2019), batch size 128, cosine decay over at most 25 epochs, gradient clipping at one and six-epoch validation patience. The original run used learning rate 3×10^{-4} in all folds. The revised run uses 6×10^{-4} in folds 1 and 2 and 3×10^{-4} in folds 3–5. Fine-tuning uses measured future increments from *training-source* curves only, source-balanced sampling with replacement,

learning rate 10^{-5} , batch size 64, at most 20 epochs and five-epoch validation patience. A pretraining checkpoint remains eligible if no fine-tuning epoch improves validation error.

The original study selected the width-256 fine-tuned model in every fold and trained three independent ensemble members. The later hyperparameter search tested 11 one-factor settings per fold. Eight architecture removals were evaluated at the full model’s selected settings.

The revised architecture decision used the mean of the five saved validation scores from the initial fits. The full model ranked first at 14.71%; removing the query residual ranked next at 14.90%. The retained design was retrained with three new independent initializations at the saved fold settings. All 15 fits finished before the revised test phase began. The same NU test partitions had already been examined, so this phase separation does not create an independent confirmatory test.

5.4 Baselines, error and interval coverage

Three prefix-only baselines use the identical day-10 readings and predict identical future times. Persistence holds the last increment constant. The log-time fit estimates one nonnegative coefficient from context. The Kelvin prefix fit estimates nonnegative coefficients of Eq. (4) from context only. These baselines do not use material descriptors. The original six-arm comparison has no property-removal or real-only initialization control; the supplementary architecture study adds a property-conditioning removal.

The error measure has three steps. First, absolute forecast errors are averaged over later readings of each curve. Second, this mean is divided by the curve’s mean absolute later response, with a minimum denominator of $1 \mu\epsilon/\text{MPa}$. Third, curve errors are averaged within each source and source averages receive equal weight. This prevents a large publication from dominating the result. For future-reading set Q_i , the curve-level scores are

$$\begin{aligned} \text{MAE}_i &= \frac{1}{|Q_i|} \sum_{j \in Q_i} |\hat{y}_{ij} - y_{ij}|, \\ \text{NMAE}_i &= \frac{\text{MAE}_i}{\max\left(|Q_i|^{-1} \sum_{j \in Q_i} |y_{ij}|, 1 \mu\epsilon/\text{MPa}\right)}. \end{aligned} \quad (2)$$

Measured future responses enter the denominator for scoring only, never as inputs. Errors are averaged across curves within a literature source and then equally across all 65 outer-test sources. This source-macro NMAE is not pointwise mean absolute percentage error. Marginal 90% coverage averages each curve’s fraction of future readings inside its central interval, first across curves within source and then equally across sources; mean interval width uses the same aggregation. Confidence intervals are percentile intervals from 2,000 bootstrap resamples of source means. Paired contrasts resample the same sources for both methods. The eight supplementary architecture intervals are exploratory and are not adjusted for multiple contrasts. These intervals condition on fitted checkpoints and selection; they do not incorporate random-initialization or cohort-selection uncertainty.

A source-bootstrap interval describes uncertainty in an aggregate error estimate. It is obtained by repeatedly resampling the 65 source averages and recomputing the score. It differs from the shaded predictive interval for an individual future reading. Neither type of interval includes the full uncertainty caused by prior model selection.

5.5 Model attribution protocol

SHapley Additive exPlanations (SHAP) attribute a model output to specified input groups relative to a background distribution (Lundberg and Lee, 2017). The analysis explains the frozen three-member ensemble’s predicted compliance increment at days 28, 90 and 160. Each eligible NU test curve is assigned only to its own outer-fold ensemble. The requested horizons are fixed across specimens. These forecasts are explanation targets, not additional observed test outcomes.

Four groups are considered: density; compressive strength; modulus together with its missingness flag; and the complete early measurement history together with first-reading time. The last group is labelled “Early history + anchor” in Figure 3. It contains the available early time-increment pairs and their first-reading time t_a . These inputs are replaced together during SHAP calculation, so a specimen’s history remains paired with its own reference time. Its attribution therefore describes their combined contribution; it does not separate the importance of the anchor from individual readings. Modulus and its missingness flag are also grouped to preserve consistent encoding.

All 16 group coalitions are evaluated. The background contains one eligible curve drawn uniformly from each training source in that fold. A second independently sampled background panel assesses sensitivity to donor selection. No validation or test curve supplies a background record. Global importance averages absolute contributions within each test source and then equally across sources. Appendix E specifies the calculation and verification checks.

6 Experimental results and ablation study

An ablation retrains the model after removing or changing one part. Comparing that model with the full version tests whether the part helps under the chosen settings. Such a comparison is conditional on the training procedure. Removing attention can also change parameter count, so it does not isolate a single causal mechanism.

6.1 Validation comparison and architecture selection

Table 4 compares prefix-only baselines, architecture variants and training strategies using validation NMAE. Scores average source-macro errors equally across five folds; lower values indicate better predictions. The Kelvin and log-time baselines fit the early readings, while persistence holds the last measured increment constant. Architecture variants remove or simplify one component, with one model fitted per fold. Parameter counts are millions per model, and parentheses give network width/layers. The small, medium and large models use 64/2, 128/4 and 256/4, respectively. Reused synthetic examples remain unchanged across training epochs; new synthetic examples are regenerated each epoch. Real-data fine-tuning subsequently adapts the pretrained model using measured training curves.

Table 4: Validation NMAE (%) for baselines, architecture selection and training controls. Scores average source-macro NMAE equally across five validation folds; lower is better. Parameters are millions per member; parentheses give width/layers. Architecture rows use initial single-model fits. Training controls use their selected checkpoints. The first four capacity controls use synthetic pretraining only. Dashes indicate that neural-network parameter counts do not apply to the baselines.

Model or change	Parameters (M)	Validation NMAE (%)
<i>Prefix-only baselines</i>		
Kelvin prefix fit	—	25.43
Log-time prefix fit	—	20.40
Persistence	—	48.36
<i>Architecture selection: one model per fold</i>		
Proposed model (full architecture)	3.10	14.71
Without context self-attention	2.05	15.33
Without query cross-attention	2.84	15.31
Without both attention paths	1.78	17.49
Without material-property input	3.10	17.06
Without query-gap feature	3.10	16.31
Without query residual path	3.10	14.90
Linear query projection	3.03	15.03
Fixed predictive scale (0.34)	3.10	15.30
<i>Capacity and training variants: one member</i>		
Small: reused synthetic examples (64/2)	0.11	18.28
Small: new synthetic examples (64/2)	0.11	18.24
Medium: new synthetic examples (128/4)	0.78	17.11
Large: new synthetic examples (256/4)	3.10	16.33
Real-fine-tuned medium (128/4)	0.78	16.15
Real-fine-tuned large (256/4)	3.10	15.18

The full architecture achieved the lowest mean validation NMAE in this comparison (14.71%), supporting its selection for CreepPFN. Removing the query residual increased the score to 14.90%; removing both attention paths increased it to 17.49%. The small model obtained similar errors with reused and new synthetic examples (18.28% and 18.24%). Larger models reduced the observed validation error, and real-

data fine-tuning reduced it further to 16.15% for the medium model and 15.18% for the large model. All learned variants had lower validation errors than the three prefix-only baselines. These results describe model selection on validation data. The 14.71% score belongs to the initial single-model fits and is not the validation score of the final ensemble. Small score differences do not establish that every retained component is essential or that the validation ranking will hold on test data. Detailed settings, paired test comparisons and ensemble results are retained in Appendix D.

6.2 SHAP attribution of the frozen forecasts

Figure 3 summarizes exact grouped SHAP values for all 610 eligible NU curves from 65 sources. Each curve is explained using its assigned outer-fold ensemble. The early-history and first-reading-anchor group has the largest source-mean absolute attribution at every horizon. Its magnitude increases from $19.78 \mu\epsilon/\text{MPa}$ at day 28 to 24.41 at day 90 and 28.84 at day 160. These values represent 93.30%, 89.97% and 88.58% of the summed absolute group attributions, respectively. These percentages are attribution shares, not explained variance or prediction accuracy.

Modulus with its missingness indicator ranks second. Its absolute attribution increases from 0.97 to 1.90 and $2.62 \mu\epsilon/\text{MPa}$ across the three horizons. Strength contributes 0.37, 0.65 and $0.85 \mu\epsilon/\text{MPa}$, whereas density contributes 0.08, 0.17 and $0.25 \mu\epsilon/\text{MPa}$. Both independently sampled background panels preserve this ranking at every horizon. The history share differs by at most 0.12 percentage points between panels. This comparison assesses sensitivity to two background samples; it is not a confidence interval.

The signed day-90 contributions show how each group shifts a specimen’s forecast relative to its fold-specific background mean. Positive values raise the forecast; negative values lower it. They do not indicate whether that forecast is accurate. All coalition decompositions passed the additive reconstruction check. Independent CPU predictions differed from the GPU full-input forecasts by at most $0.0057 \mu\epsilon/\text{MPa}$ and 0.0028% in relative terms.

The prominence of measured history is consistent with the value of specimen observations in previous concrete creep research. A Bayesian updating study combined creep-model distributions with in-situ specimen measurements (Sousa et al., 2018). Sequential updating reduced estimated prediction variability for the structure examined. That study supports using observations to refine creep predictions. It does not validate the present day-10 cutoff or the numerical SHAP shares, because its monitoring period and updating method differ.

Previous SHAP results also show why the present ranking requires a task-specific interpretation. An ensemble-learning study identified time since loading, compressive strength, loading age, relative humidity and temperature as the five most influential inputs in its concrete creep-compliance models (Liang et al., 2022). Here, the forecast horizon is held fixed during attribution, and the predictor receives an observed response history. Loading age, humidity and temperature are not included as descriptors. Their effects may be partly reflected in the early history, but this remains an interpretation rather than a measured decomposition. The smaller strength attribution therefore does not contradict its importance in models with different inputs and prediction tasks.

Material descriptors remain relevant despite their smaller direct attributions. The Model B4 calibration study linked creep-model parameters to concrete composition and strength (Wan-Wendner et al., 2015). This provides precedent for material-dependent curve parameters, as used in the present generator. It does not establish the specific ordering of modulus, strength and density found here. In particular, the increasing modulus attribution indicates sensitivity of the fitted predictor; it does not demonstrate a causal increase in the influence of stiffness with forecast time.

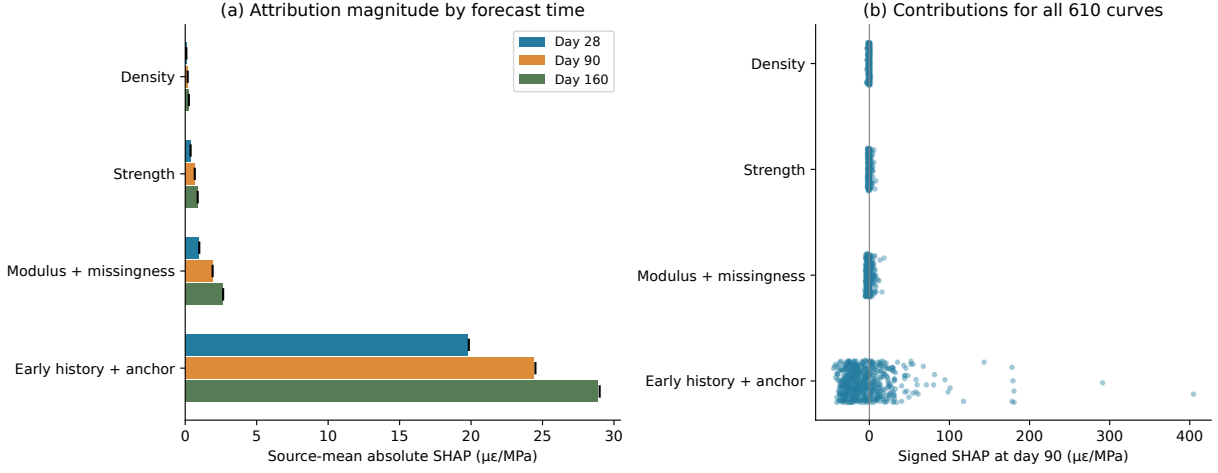


Figure 3: Grouped SHAP analysis of the frozen outer-fold ensembles on 610 NU curves. (a) Absolute contributions are averaged within each source and then equally across 65 sources. Bars use the first training-only background panel; black ticks use the independently sampled second panel. (b) Each point is one curve’s signed contribution at day 90 using the first panel. This panel displays individual curves without source weighting. Early history includes reading times, responses and the first-reading anchor. Modulus includes its missingness indicator. Contributions explain model predictions relative to the selected backgrounds, not causal material effects.

7 Practical application and limits

7.1 Using CreepPFN for a specimen

Application begins with the material record and at least two measurements through day 10. Reading times must be expressed as elapsed days since loading. Compliance is referenced to the first recorded reading using Eq. (1). The supplied modulus must be recorded independently of the future curve. If it is unavailable, the missingness flag and training-derived imputation are used.

The user then specifies future times after day 10 and within the studied 160-day horizon. The frozen ensemble returns a point forecast and a central 90% marginal interval at each time. Model weights are not updated using the new specimen’s later readings. For NU evaluation, each specimen is forecast by its assigned outer-fold model. The external study applies all five frozen fold ensembles and reports their performance separately before averaging scores.

The output can support interpretation of the early test and planning of subsequent measurements. Those uses remain proposed applications; operational benefit was not measured. The predicted increment must not be treated as a complete structural deformation estimate. Such an estimate also requires the stress history, geometry and other deformation components. The present study does not validate structural design decisions.

7.2 Application to KMUTT and other-literature records

After revised NU fitting and evaluation, each fold’s three trained members were frozen and applied to the same 66 external curves. Each fold ensemble predicts every curve once. The five scores are correlated descriptions of trained models rather than independent external samples. The primary external score first calculates each curve’s later-reading MAE divided by $\max\{\text{mean } |y_{\text{future}}|, 1 \mu\epsilon/\text{MPa}\}$ and then averages equally across curves. Table 5 reports the mean of five fold scores and their range. This curve-macro NMAE differs from NU source-macro NMAE because one KMUTT laboratory and one unresolved literature label do not provide enough identified independent sources for source-level inference. These external records were examined in the historical study. The revised transfer results are therefore descriptive reassessments.

Table 5: External day-10 forecasts from five frozen NU-trained three-member ensembles. Each model tests the same curves and later readings. CreepPFN entries are mean fold curve-macro NMAE with the descriptive fold range in brackets. Prefix baselines use only each curve’s day-10 context and have the same score in every fold. Values are percentages; lower is better. The primary CreepPFN test supplies recorded E_{rec} as an unverified proxy for the E_{28} input. The literature result assumes a nominal applied stress of $0.4f_c$ because curve-specific stress is not identified.

Origin, curves/readings	CreepPFN	Log-time	Kelvin	Persistence
KMUTT, 41/569	10.39 [9.20, 11.93]	19.63	29.22	45.50
Other literature*, 25/276	20.71 [18.16, 22.58]	21.07	26.75	53.30
Pooled, 66/845	14.30 [13.31, 15.92]	20.18	28.29	48.46

*Day 10 is the context cutoff. All 25 other-literature curves have their last available early reading on day 7 and their first future reading on day 14. No day-10 reading is available or interpolated.

On the KMUTT curves, mean fold curve-macro MAE was $4.08 \mu\epsilon/\text{MPa}$, with 98.72% curve-macro coverage of the nominal 90% marginal interval and mean interval width $24.18 \mu\epsilon/\text{MPa}$. On the unresolved literature set, these values were $6.07 \mu\epsilon/\text{MPa}$, 88.11% and $22.75 \mu\epsilon/\text{MPa}$. The scores use the specified response conversion. They indicate lower empirical error for KMUTT than for the other-literature set under that convention. Curve-specific publications and applied stresses remain unresolved for the latter cohort. In the matched missing-modulus sensitivity test, mean fold curve-macro NMAE was 11.70% for KMUTT, 18.00% for other literature and 14.09% pooled. Supplying recorded modulus improved KMUTT error but worsened other-literature error. The contrast cannot identify a causal effect of true 28-day stiffness because measurement age and source conditions are unresolved.

A curve-level audit confirmed that all 25 literature curves and their 276 post-day-10 readings entered every revised fold evaluation. Their first recorded elapsed day t_a is zero; the separate concrete age-at-loading field is blank for all 25 metadata rows. These limitations preclude attribution of heterogeneous errors to publications or loading programmes.

7.3 Examples across NU and external origins

Figure 4 presents six NU examples and the first curve from each external cohort. The NU examples use six evenly spaced source identifiers. The external examples use the revised fold-1 ensemble. Selection is independent of forecast error.

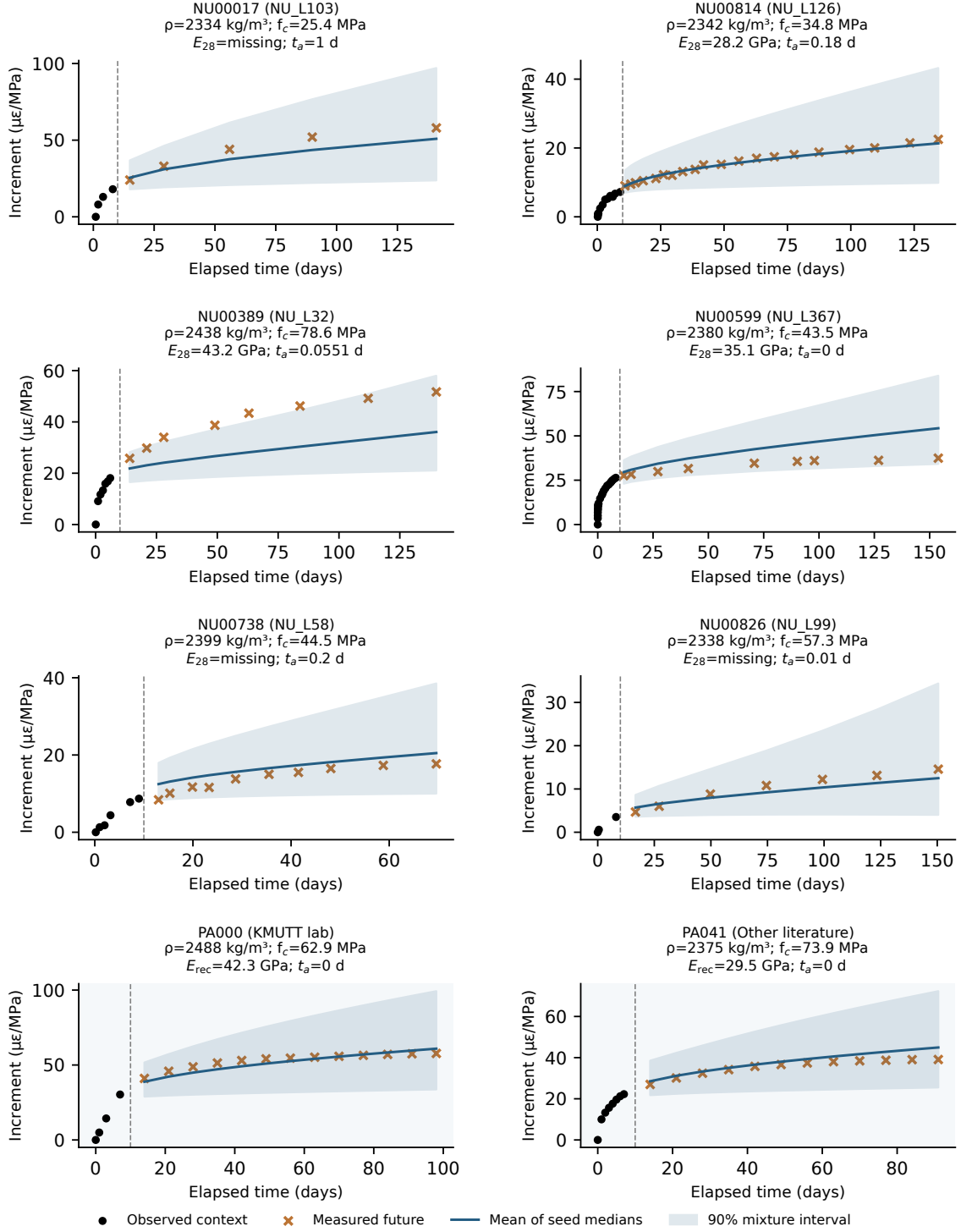


Figure 4: Eight forecasts from the revised CreepPFN. The first six panels represent NU sources. The last two show KMUTT and other-literature examples under the nominal $0.4f_c$ response conversion. Panel headings give material inputs and first-reading time. External modulus is an unverified E_{28} proxy. The blue line averages member medians; shading is the central 90% predictive mixture interval. Axes differ between panels.

7.4 Interpreting an individual forecast

The black points in Figure 4 are the readings supplied to the model. The dashed vertical line marks the end of the observation window. The blue curve shows the ensemble forecast at later query times. Orange crosses are the measurements used only to assess the forecast. Shading describes uncertainty at each time. It is not a physical boundary that the experiment must obey.

The final two panels illustrate transfer to KMUTT and the other-literature cohort. Their material annotations show recorded external modulus rather than verified 28-day modulus. A value of $t_a = 0$ identifies the first elapsed reading day. These eight examples are selected by identifiers rather than low forecast error. Cohort accuracy must therefore be read from the result tables rather than inferred from the displayed curves.

8 Discussion

8.1 Interpretation of the prediction and ablation results

The NU results support the feasibility of forecasting later creep from material descriptors and an early measured history. The revised CreepPFN attained 17.40% source-macro NMAE, compared with 26.51% for the Kelvin prefix fit on the same evaluation cohort (Appendix D). This comparison concerns extrapolation within the studied time horizon and source partitions. It does not establish accuracy under arbitrary loading histories or environmental conditions.

The architecture results suggest that access to the measured history and the extrapolation gap contributes to prediction. Removing both attention paths increased NMAE by 1.69 percentage points. Removing the query-gap feature increased it by 0.86 points (Table 8). However, removing both attention paths also reduced model capacity. The contrast therefore does not isolate attention from parameter count. Several other ablation intervals included zero. Those results indicate unresolved effects under the tested design, rather than proof that the corresponding components have no value.

Validation selection did not guarantee improved performance after retraining. The revised model exceeded the historical selected ensemble’s error by 0.60 percentage points, with a paired source-bootstrap interval of 0.17–1.05 points. Its difference from the earlier tuned full model was smaller and uncertain. Model selection and random initialization therefore require separate consideration. The reported comparisons do not establish that increasing network size alone would improve generalization. They also do not isolate the benefit of synthetic pretraining, because a matched model trained only on real curves from random initialization was not evaluated.

8.2 Interpretation of the SHAP analysis

The grouped attribution results indicate that the frozen predictor distinguishes specimens mainly through their early measured histories under the chosen background distribution (Figure 3). This is consistent with the model’s use of early responses to encode specimen behaviour and determine response scale. The increasing modulus attribution at longer horizons suggests greater model sensitivity to this descriptor as extrapolation extends. This observation concerns model behaviour and does not establish a physical mechanism.

The history group contains multiple measurements and their times, whereas density and strength are individual descriptors. Their attribution magnitudes therefore do not provide an equal-sized comparison of physical variables. Early responses may already reflect material and environmental influences. Material relationships also enter through the synthetic training generator. Small direct property attributions cannot establish that those properties are physically unimportant or that material-conditioned training is unnecessary.

SHAP and retrained ablations address different questions. SHAP describes contributions within a fixed fitted model. An ablation measures performance after an input or component is removed and the model is retrained. Agreement between two background panels supports limited sampling stability, but not robustness to other attribution definitions. Replacing groups can combine properties and histories that are uncommon in the experimental data. The resulting interventional attributions are reference-dependent and noncausal. Conditional attribution methods and additional background samples would help assess this limitation. This analysis concerns NU forecasts and does not explain the external literature cohort’s larger errors.

8.3 External performance and material information

External curve-macro NMAE was 10.39% for KMUTT and 20.71% for the other-literature set (Table 5). These scores describe different collections of specimens and measurement schedules. They cannot be

compared directly with NU source-macro NMAE as if all three used the same sampling and weighting scheme. The lower KMUTT score does not establish broader generalization from one laboratory.

The modulus sensitivity analysis also differed between cohorts. Treating modulus as missing increased KMUTT error to 11.70%, but reduced other-literature error to 18.00%. This observation shows that supplying an additional recorded descriptor does not necessarily improve transfer. Possible explanations include differences in modulus measurement age, material combinations or source conditions. These explanations remain hypotheses. The experiment does not show that modulus is physically unimportant or that the recorded values are incorrect. Choosing whether to omit it after examining external errors would introduce further selection on the evaluation cohort.

8.4 Limitations of the present evidence

Source-disjoint partitions reduce direct sharing of publication-specific information within each NU fold. Nevertheless, the same outer sources were inspected in earlier experiments before the revised architecture was selected. The present results are therefore a reassessment, rather than a new independent confirmation. The source-bootstrap intervals condition on the fitted models and observed sources. They do not include the full uncertainty from architecture selection, training randomness or cohort construction.

External provenance remains incomplete. The 25 literature curves lack specimen-level publication mappings and verified applied stresses. Their response conversion assumes a stress of $0.4f_c$, and recorded modulus enters as an unverified 28-day proxy. Agreement between earlier and later spreadsheet values cannot validate those physical assumptions. Missing loading age, exposure conditions and mixture composition further limit interpretation. Synthetic tasks expand the training distribution but do not add independent experimental sources or recover unrecorded material variables.

The generator represents a restricted family of increasing creep responses. It omits unloading, varying environments, cyclic loading and accelerating tertiary creep. The predictor does not enforce nonnegative increments or monotonic trajectories. Its uncertainty intervals are marginal at each query time. NU coverage of 94.51% for a nominal 90% interval, together with the reported interval width, does not establish calibration for every material or horizon. A useful interval must balance coverage and precision; aggregate coverage alone cannot demonstrate engineering reliability.

8.5 Priorities for further research

The first priority is a publication-resolved external dataset. Each curve should retain its specimen identifier, actual applied stress, loading age, modulus measurement age, exposure conditions and response definition. A new laboratory or publication cohort should then be reserved before model development. Architecture, preprocessing and checkpoint-selection rules should be fixed before that cohort is evaluated.

A second priority is to separate the contributions of the generator and predictor. Controlled studies should compare real-only training, synthetic pretraining without real-data adaptation, and the combined procedure under comparable computational budgets. Material-conditioned and unconditioned generators should be compared independently of the network’s property inputs. Attention comparisons with similar parameter counts and repeated independent fits would help distinguish architectural effects from capacity and training variability.

Further model development should evaluate loading and environmental descriptors when reliable records are available. Broader generators could represent additional response regimes, with validity checked against held-out experiments. Physical output constraints should be tested for both their accuracy benefits and their potential to suppress legitimate measurement variation. Uncertainty assessment should report calibration and interval width by source, material class and forecast horizon. Prospective laboratory studies are finally needed to determine whether forecasts improve monitoring or test-planning decisions. These directions are proposed extensions, not demonstrated capabilities of the present model.

9 Conclusions

CreepPFN provides a material-conditioned prior-data fitted framework for concrete creep forecasting. A generator calibrated on real training sources produces synthetic time–response tasks. An attention network learns from these tasks and is subsequently fine-tuned on measured training curves. For a new specimen,

the frozen model uses material descriptors and available readings through day 10 to predict later compliance increments and marginal uncertainty intervals. No specimen-specific weight update is required.

Five source-disjoint NU folds included 610 evaluable curves from 65 literature sources. The revised three-member ensembles attained 17.40% source-macro NMAE, with a 95% source-bootstrap interval of 14.96–20.14%. The Kelvin prefix baseline attained 26.51% on the same cohort. The historical selected ensemble attained 16.80%, so revised retraining did not improve accuracy. This result distinguishes the validation-selected configuration from the best observed historical fit. The ablations supported the combined attention paths and the query-gap feature under the tested settings. They did not establish the separate benefit of every component or isolate synthetic pretraining from real-data adaptation.

Frozen external evaluations yielded mean fold curve-macro NMAE of 10.39% on 41 KMUTT curves and 20.71% on 25 other-literature curves. The difference indicates that performance depends on the evaluation cohort. Unresolved publication mappings, assumed loading stress and uncertain modulus measurement age restrict interpretation of these external results. The nominal 90% intervals achieved 94.51% source-macro coverage on NU, but this aggregate result does not establish calibration for individual materials or complete trajectories.

The principal contribution is the integration of a material-linked task generator with early-measurement-conditioned creep forecasting. The present evidence supports further evaluation of this approach within the studied response regime and time horizon. Reuse of previously examined test cohorts limits independent confirmation. Future work should prioritize verified experimental metadata, prospectively reserved sources, matched real-only training controls and conditional uncertainty assessment. These steps are necessary to establish the method’s reliability for practical monitoring and test planning.

Data, code and reproducibility

The supplementary CreepPFN package contains processed fold data, training-only priors, all 15 final checkpoints, training code and software tests. It provides batch prediction through the `creep-pfn predict` command and an interactive interface through `creep-pfn app`. Both accept density, strength, optional recorded modulus and at least two compliance readings through day 10. A packaged manifest records hashes for the frozen checkpoints, priors, reported results and manuscript. The application can use one evaluated three-member fold ensemble or pool all 15 members. The pooled option is labelled as an exploratory deployment ensemble because it is not the unit used to calculate the reported cross-validation metric. No public archival deposition is claimed.

The revised GPU run used PyTorch 2.5.1+cu121, NumPy 1.26.4, pandas 2.2.2, SciPy 1.14.1 and scikit-learn 1.5.2 on an RTX 4090. Its runner is `python -m creep_prior.revised_main`. Training and test phases are separate. The local `CV_PROTOCOL.md`, `TRAINING.md` and `ARCHITECTURE_STUDY.md` record the underlying procedures. This manuscript uses the same fitted results as `paper3.pdf`. The new figures are regenerated with `python -m manuscript.make_paper4_figures`; input hashes and counts are stored in `manuscript/paper4_figure_audit.json`.

The SHAP runner is `python -m creep_prior.shap_study`. Saved coalition values, background identifiers, checkpoint hashes and attribution summaries are in `artifacts/shap_study`. The protocol is recorded in `SHAP_PROTOCOL.md`. Exact enumeration of all 16 group coalitions for 610 curves, three horizons and two background panels took 41.2 seconds on a rented RTX 4090, excluding environment setup and file transfer. Figure 3 is regenerated with `python -m manuscript.make_shap_figure`. Cross-device prediction checks use relative tolerance 5×10^{-5} and absolute tolerance $2 \times 10^{-4} \mu\epsilon/\text{MPa}$; the original stricter relative tolerance of 2×10^{-5} flagged 25 of 1,830 comparisons. The measured maximum differences are reported in Section 6.2.

A Synthetic generator specification

A retardation-spectrum representation motivates response components with different characteristic times (Bažant and Xi, 1995). The present generator uses two restricted, increasing and concave latent curve families as a task distribution; it does not implement solidification theory or a complete constitutive law. For first-reading elapsed day t_a and day-160 magnitude $M > 0$, the anchor-normalized Weibull family is

$$g_W(t) = M \frac{\exp[-b(t_a/160)^c] - \exp[-b(t/160)^c]}{\exp[-b(t_a/160)^c] - \exp(-b)}, \quad b > 0, \quad 0.1 < c < 1. \quad (3)$$

The Kelvin-plus-log family is

$$\begin{aligned} \phi_k(t) &= 1 - \exp(-t/\tau_k), \quad \tau_k \in \{0.3, 3, 30, 300\} \text{ days}, \\ \phi_5(t) &= \log(1 + t/10), \quad h_k(t; t_a) = \frac{\phi_k(t) - \phi_k(t_a)}{\phi_k(160) - \phi_k(t_a)}, \\ g_K(t) &= \sum_{k=1}^5 A_k h_k(t; t_a), \quad A_k \geq 0, \quad M = \sum_{k=1}^5 A_k. \end{aligned} \quad (4)$$

Both families are fitted to each complete *training-source* curve at its recorded observation times. The Weibull fit uses three exponent starts and bounded nonlinear least squares. The Kelvin-plus-log fit uses nonnegative least squares with a weak amplitude penalty.

The raw descriptor vector is

$$\mathbf{x}_i^{\text{raw}} = [\log \rho_i, \log f_{c,i}, \log E_{28,i}, \mathbf{1}(E_{28,i} \text{ missing}), \log(1 + t_{a,i})]^\top. \quad (5)$$

Missing log modulus is replaced by the fold-training median; each feature is centered and scaled with training-only statistics. For each family, fitted parameter vectors are transformed to an unconstrained scale and related to properties by

$$\mathbf{z}_i = \boldsymbol{\beta}_0 + B\mathbf{x}_i + \mathbf{u}_{s(i)} + \boldsymbol{\epsilon}_i. \quad (6)$$

Ridge regression uses inverse-source-size weights. Grouped cross-validation *inside the training groups* selects its penalty from $\{1, 10, 100, 1000\}$ separately for each family and fold. Centered whole-vector source and within-source residuals are sampled empirically to preserve parameter-component dependence. Sixty-four coefficient matrices per fold are obtained by bootstrap resampling training sources. The resulting generator is data-informed and empirical; it is not an identified Bayesian posterior over constitutive parameters.

A donor training source supplies an observed combination of descriptors and an irregular time schedule. Synthetic magnitude and shape parameters are sampled conditionally, with equal design probabilities for the two families, and transformed parameter draws are clipped to an expanded training support. Reading discrepancy is estimated from the training fit residual relative to M and used to generate independent reading errors e_j before anchor subtraction:

$$\tilde{y}_j = g(t_j) + e_j - e_a, \quad e_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_e^2). \quad (7)$$

The shared anchor error induces dependence between future increments. A prefix length is sampled from two through one less than the number of donor readings, and subsequent readings become synthetic query targets. Synthetic family, latent parameters, donor identifier and future target responses are excluded from predictor inputs.

B Network inputs and forward equations

For observed context $\mathcal{C} = \{(t_j, y_j) : j \leq n_c\}$, standardized descriptors $\mathbf{x}$ and a requested time t_* , CreepPFN estimates a marginal response distribution. A context-only response scale

$$s = \max \left(\max_{j \leq n_c} |y_j|, 1 \mu\text{E}/\text{MPa} \right) \quad (8)$$

normalizes each task. Context tokens contain $\log(1 + t_j)/\log(161)$, $t_j/160$ and $\text{asinh}(y_j/s)$. A property network encodes $[\mathbf{x}, \log(1 + s)]$ and adds its embedding to each context token. The encoder uses pre-normalized transformer layers and GELU activation (Vaswani et al., 2017). Dropout is 0.05 in folds 1, 2,

4 and 5, and zero in fold 3. Masked padding is sanitized before summary statistics or embeddings are calculated.

Each future query represents linear and log time, log time since the last context observation and the transformed last response.

Let $\mathbf{c}_j$ be a three-feature context token before projection and $\mathbf{d}_*$ a four-feature query token. The forward map can be summarized as

$$\begin{aligned} \mathbf{p} &= P_{\text{prop}}([\mathbf{x}, \log(1+s)]), \quad H = \text{LN}(\text{Enc}_4(\{P_c(\mathbf{c}_j) + \mathbf{p}\}_{j \leq n_c})), \\ \mathbf{q}_* &= P_q(\mathbf{d}_*) + \mathbf{p}, \quad (r_*, a_*) = \text{Head}[\mathbf{q}_* + \text{Attn}(\mathbf{q}_*, H, H)], \\ \mu_* &= \text{asinh}(y_{n_c}/s) + r_*, \quad \sigma_* = \text{softplus}(a_*) + 0.03. \end{aligned} \quad (9)$$

Here P_{prop} is the property network, P_c and P_q are context and query projections, Enc_4 is the four-layer context encoder, and Attn is cross-attention with context keys and values. The padding mask excludes absent context readings. Distinct future queries share the encoded context but do not exchange information or recursively feed predictions into later queries.

C Training objective and predictive intervals

For $z = \text{asinh}(y/s)$, training minimizes task-balanced Gaussian negative log likelihood,

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{1}{|Q_i|} \sum_{j \in Q_i} \left[\frac{(z_{ij} - \mu_{ij})^2}{2\sigma_{ij}^2} + \log \sigma_{ij} + \frac{1}{2} \log(2\pi) \right]. \quad (10)$$

The transformation Jacobian is independent of predictor parameters and omitted from optimization. A member’s response median and central 90% marginal interval are

$$\hat{y} = s \sinh(\mu), \quad [L, U] = s \sinh(\mu \pm 1.6448536\sigma). \quad (11)$$

For three members with the same context scale s , the final point forecast is $s \sum_{m=1}^3 \sinh(\mu_m)/3$. The ensemble interval uses the equal-weight mixture cumulative distribution

$$F(y) = \frac{1}{3} \sum_{m=1}^3 \Phi\left(\frac{\text{asinh}(y/s) - \mu_m}{\sigma_m}\right), \quad [L_{\text{ens}}, U_{\text{ens}}] = [F^{-1}(0.05), F^{-1}(0.95)]. \quad (12)$$

Here Φ is the standard normal cumulative distribution. Mixture quantiles are obtained by numerical bisection. They are not averages of member interval endpoints. These distributions are marginal at separate query times, rather than simultaneous bands for a complete trajectory.

D Detailed ablation comparisons and validation search

The revised full ensemble obtained 15.44% validation NMAE and 17.40% test source-macro NMAE (95% source-bootstrap interval 14.96–20.14%). Test MAE was $6.53 \mu\epsilon/\text{MPa}$, with 94.51% coverage and mean width $31.73 \mu\epsilon/\text{MPa}$ for the central 90% predictive intervals. The earlier validation-tuned ensemble obtained 15.05% validation and 17.28% test NMAE. The earlier selected ensemble obtained 15.29% validation and 16.80% test NMAE. Validation scores average fold-level source-macro scores equally; test scores weight the 65 sources equally. The revised-minus-earlier-selected test difference was +0.60 percentage points (95% paired interval 0.17–1.05). The revised-minus-earlier-tuned difference was +0.12 points (−0.42 to 0.68). These results do not establish an accuracy improvement from revised retraining or ensembling. Earlier inspection of the same test partitions makes the reassessment exploratory.

The tables below retain matched training contrasts, fold-specific tuning and architecture contrasts. The architecture reference and its eight variants use matched random initializations. It differs from the proposed model retrained with three new independent initializations.

The small model has width 64 and two layers (113,986 parameters). The medium model has width 128 and four layers (779,906 parameters). The large model has width 256 and four layers (3,099,906 parameters). The six historical training arms use a common initialization setting, five folds and task count.

The 11-candidate search changes one factor at a time: width (128 or 256), depth (2, 4 or 6), heads (4 or 8), dropout (0, 0.05 or 0.15), synthetic learning rate (10^{-4} , 3×10^{-4} or 6×10^{-4}), or fine-tuning learning rate (3×10^{-6} , 10^{-5} or 3×10^{-5}). The lowest initial-fit validation NMAE selects the fold setting. Two additional independent fits repeat that setting. This bounded search is not exhaustive.

Context self-attention removal retains residual token-wise feed-forward layers. Cross-attention removal substitutes a padding-masked context mean. The linear-query control replaces the multilayer projection with one linear layer. The fixed-scale control retains Gaussian NLL with transformed-space scale 0.34. All architecture arms use independent validation checkpoint selection and the same three-member aggregation.

Table 6: Paired source-macro NMAE contrasts in percentage points. Differences are first method minus reference; negative values favor the first method. All 65 sources contribute. Revised-versus-baseline contrasts use the frozen revised predictions and the same baseline forecasts; remaining training contrasts come from the historical single-model study.

Contrast	Difference	95% paired interval
Small: new – reused examples	−0.20	[−1.66, 1.19]
New examples: medium – small	−0.84	[−1.88, 0.30]
New examples: large – medium	−0.22	[−1.50, 0.98]
Medium: real fine-tuning – synthetic only	−1.57	[−2.67, −0.67]
Large: real fine-tuning – synthetic only	−2.00	[−3.76, −0.58]
Revised ensemble – small, reused examples	−2.88	[−4.49, −1.41]
Revised ensemble – Kelvin prefix	−9.11	[−12.25, −6.05]
Revised ensemble – log-time prefix	−9.37	[−23.31, −0.54]
Revised ensemble – persistence	−29.86	[−33.18, −26.11]

Table 7: Fold-specific tuning on separate NU validation sources. Validation columns are initial-fit source-macro NMAE. The last column is the three-member selected full model on that fold’s outer-test sources. All values are percentages. The pooled 65-source score is computed across sources, not by equally averaging the five fold scores.

Outer fold	Selected setting	Reference validation	Selected validation	Tuned outer test
01	Pretrain LR 6×10^{-4}	12.94	12.27	18.74
02	Pretrain LR 6×10^{-4}	15.42	14.74	17.65
03	Dropout 0	17.48	16.97	14.00
04	Reference	15.45	15.45	18.45
05	Reference	14.14	14.14	17.66

Table 8: Three-member nested architecture ablations on the same 610 curves, 65 source-disjoint groups and 5,646 future readings. Parameter counts are millions for each selected fold configuration. NMAE and its 95% source-bootstrap interval are percentages. Paired differences are removal minus tuned full model in percentage points; positive values indicate higher error. Paired intervals resample matched sources.

Architecture arm	Parameters (M)	NMAE [95% interval]	Removal–full [95% interval]
Tuned full model	3.10	17.28 [14.80, 19.94]	Reference
No context self-attention	2.05	16.99 [14.64, 19.48]	−0.29 [−0.81, +0.20]
No query cross-attention	2.84	17.05 [14.87, 19.40]	−0.23 [−0.99, +0.42]
No attention	1.78	18.97 [16.39, 21.81]	+1.69 [+0.34, +3.13]
No property conditioning	3.10	17.82 [15.27, 20.91]	+0.54 [−1.45, +2.51]
No query-gap feature	3.10	18.14 [15.84, 20.64]	+0.86 [+0.27, +1.46]
No query residual path	3.10	16.71 [14.24, 19.41]	−0.57 [−1.23, +0.07]
Linear query projection	3.03	17.17 [14.65, 20.09]	−0.11 [−0.83, +0.59]
Fixed predictive scale	3.10	17.25 [14.68, 20.24]	−0.04 [−0.70, +0.60]

The validation scores below document the bounded candidate search. The interval statistics describe the earlier architecture ensembles. They are separate from the revised main model from independent retraining.

Table 9: Complete bounded one-factor search: initial-fit source-macro validation NMAE (%). Each fold uses its own disjoint validation sources. Bold values mark the setting selected within that fold. Candidate rows differ in one factor from the reference. These validation scores are not outer-test results.

Candidate	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Reference	12.94	15.42	17.48	15.45	14.14
Width 128	13.17	17.29	19.31	17.41	15.54
Two layers	14.45	15.33	17.20	16.28	15.01
Six layers	12.96	15.50	18.16	16.67	17.14
Eight heads	13.09	15.55	17.90	16.42	15.04
Dropout 0	13.14	15.01	16.97	16.95	15.19
Dropout 0.15	12.88	14.99	18.30	16.28	14.99
Pretrain LR 10^{-4}	13.26	17.16	19.12	16.76	14.92
Pretrain LR 6×10^{-4}	12.27	14.74	17.52	15.87	14.27
Fine-tune LR 3×10^{-6}	13.13	15.78	17.51	15.57	14.57
Fine-tune LR 3×10^{-5}	12.32	15.68	17.58	15.56	14.50

Table 10: Descriptive accuracy and interval behavior of the nine nested three-member ensembles on the common outer-test sources. MAE and mean marginal interval width use $\mu\epsilon/\text{MPa}$. Coverage is the source-macro mean fraction of future readings inside central 90% intervals. These are fixed-checkpoint estimates; no interval for coverage or width is reported.

Architecture arm	Source-macro MAE	Coverage (%)	Mean width
Tuned full model	6.31	92.30	29.24
No context self-attention	6.47	93.97	29.62
No query cross-attention	6.50	94.05	32.72
No attention	6.81	91.38	31.95
No property conditioning	7.16	96.37	42.98
No query-gap feature	6.61	91.82	29.14
No query residual path	6.18	94.31	30.74
Linear query projection	6.36	90.47	27.44
Fixed predictive scale	6.27	97.74	51.41

E Grouped SHAP calculation

Let G contain the four input groups and let f_h be the frozen ensemble forecast at horizon h . For a target input x , a coalition $S \subseteq G$ retains groups in S and replaces the others with a training-background donor b . The finite-background game and contribution of group j are

$$\begin{aligned} v_h(S; x) &= \frac{1}{B} \sum_{b=1}^B f_h(x_S, b_{G \setminus S}), \\ \phi_{j,h}(x) &= \sum_{S \subseteq G \setminus \{j\}} \frac{|S|!(3 - |S|)!}{4!} [v_h(S \cup \{j\}; x) - v_h(S; x)]. \end{aligned} \quad (13)$$

Omitted groups are taken jointly from the same donor. Each panel contains one donor per eligible training source, with equal weights. Sampling is reproducible and performed separately for each fold and background panel. The stored fold-training scaler transforms all descriptors. Query times remain fixed when history changes. The unmasked prediction is checked against direct ensemble inference, and the additive identity $f_h(x) = v_h(\emptyset; x) + \sum_j \phi_{j,h}(x)$ is checked numerically.

Enumeration is exact for the chosen four-group game and finite background. It does not estimate conditional expectations under the full joint material distribution. Combining retained and donor groups can produce combinations absent from the experimental data. Attributions therefore describe model behavior rather than causal material effects. The history contribution includes its timing, magnitude, shape and anchor descriptor; it does not resolve individual readings. The modulus contribution includes missingness. Neither is directly comparable to removing only a single scalar input.

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